

7.2.2 Software Configuration Management Process

NOTE — The Software Configuration Management Process is a specialization of the **Configuration Management Process** from the Project Process Group in this International Standard.

7.2.2.1 Purpose

The purpose of the Software Configuration Management (SCM) Process is to **establish and maintain the integrity of software items** throughout the life cycle and to make controlled versions and releases available to authorized parties in a timely and traceable manner.

7.2.2.2 Outcomes

As a result of successful implementation of the Software Configuration Management Process:

- a) a software configuration management strategy is developed;
 - b) items generated by the process or project are identified, defined, and baselined;
 - c) modifications and releases of the items are controlled;
 - d) modifications and releases are made available to affected parties;
 - e) the status of the items and modifications are recorded and reported;
 - f) the completeness and consistency of the items are ensured; and
 - g) the storage, handling, and delivery of the items are controlled.
-

7.2.2.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Configuration Management Process.

7.2.2.3.1 Process Implementation

This activity consists of the following task:

7.2.2.3.1.1

A **Software Configuration Management Plan (SCMP)** shall be developed. The plan shall describe:

- SCM activities, responsibilities, procedures, and schedule;
- organizational roles and interfaces (e.g., system CM, suppliers, subcontractors);
- configuration item (CI) identification scheme;
- change control procedures and escalation paths;
- configuration status accounting and evaluation procedures;
- release and delivery policies and tools used (e.g., version control systems, CMDB).

The plan shall be documented, approved, and implemented.

NOTE — The SCMP may be a part of the system configuration management plan.

7.2.2.3.2 Configuration Identification

This activity consists of the following task:

7.2.2.3.2.1

A **scheme shall be established for identification of software items and their versions** to be controlled. For each item and version, the following shall be identified:

- unique item identifiers;
- documentation that establishes the baseline;
- version references and labeling conventions;
- classification (e.g., safety-critical, security-sensitive, standard);
- relationships to other configuration items (e.g., parent-child, dependencies);
- associated metadata (author, date, branch, environment).

Baselines shall be defined at appropriate project milestones (e.g., development, integration, qualification, release) and approved prior to use.

7.2.2.3.3 Configuration Control

This activity consists of the following task:

7.2.2.3.3.1

The project shall implement structured **change control** procedures, including:

- identification and recording of change requests (CRs);
- impact analysis (technical, cost, schedule, risk);
- review and decision by the Change Control Board (CCB) or equivalent authority;

- approval or disapproval of the change;
- implementation of approved changes;
- verification of the implementation and regression testing as applicable;
- formal closure of the change record.

The **Software Problem Resolution Process (7.2.8)** may provide support for this activity.

7.2.2.3.4 Configuration Status Accounting

This activity consists of the following task:

7.2.2.3.4.1

The **status of configuration items, change requests, baselines, and releases** shall be recorded and reported. Status accounting shall provide timely and accurate information about:

- CI versions and their relationships;
- current baseline configurations;
- outstanding and completed changes;
- release history;
- traceability between CIs and documentation.

Configuration status records shall be accessible to authorized parties and maintained in the project's configuration management repository.

7.2.2.3.5 Configuration Evaluation

This activity consists of the following task:

7.2.2.3.5.1

Configuration evaluations shall determine and ensure:

- the **functional completeness** of software items against their requirements;

- the **physical completeness** of software items (i.e., design and code accurately reflect current technical descriptions);
- conformance to the approved baselines;
- resolution of discrepancies identified in prior evaluations or audits.

Periodic configuration audits shall be performed to maintain integrity and traceability.

7.2.2.3.6 Release Management and Delivery

This activity consists of the following task:

7.2.2.3.6.1

The **release and delivery** of software products and documentation shall be formally controlled. Specifically:

- master copies of code and documentation shall be maintained for the lifetime of the software product;
 - release packages shall be uniquely identified, documented, and traceable to their baseline;
 - safety- or security-critical items shall be handled, stored, packaged, and delivered in accordance with organizational policies;
 - digital signatures or checksums shall be applied to ensure integrity;
 - delivery records and rollback information shall be maintained.
-

7.2.2.4 Deliverables

The following deliverables shall be produced during this process:

- Software Configuration Management Plan (SCMP);
 - Configuration Item Identification Scheme;
 - Baseline records and configuration item lists;
 - Change Requests (CR) and associated impact analyses;
 - Configuration Status Reports and Logs;
 - Configuration Evaluation Reports and Audit Records;
 - Release packages and delivery records;
 - Digital signatures or integrity check artifacts;
 - SCM tool configuration and access control records.
-

7.2.2.5 Audit and Configuration Control

All SCM artifacts (plans, baselines, change records, status logs, releases) shall be placed under configuration management themselves.

Audit evidence shall be retained for at least one release cycle or per regulatory requirements.

Regular **configuration audits** shall ensure traceability, baseline integrity, and change compliance.

Access to SCM repositories shall be controlled, logged, and periodically reviewed.